Wi-Fi Connectivity Lab

ping | traceroute | arp -a | report

Name: _____ Class: _____ Date: _____

This guide walks you through every task in simple steps, explains what each result means, and gives you a ready-made report page. Because these tests run on **your own devices**, the outputs shown here are **samples**. Run the commands yourself and write your real results in the boxes provided.

Quick vocabulary. IP address = a device's address on the network. **Ping** = sends a small test message and waits for a reply. **Hop** = one router your data passes through. **ARP** = a table that links IP addresses to physical (MAC) addresses.

1 Ping Between Your Laptop and a Friend's Phone

1	**Join the same Wi-Fi** Connect your laptop and your friend's phone to the **same** network (same name, not a guest network).
2	**Find the phone's IP** On the phone: **Settings > Wi-Fi > tap the connected network > IP address** (e.g. 192.168.1.15). Menu names vary a little by brand.
3	**Find your laptop's IP** Open Command Prompt and type `ipconfig`. Note the **IPv4 Address**. The first three numbers should match the phone's.
4	**Ping the phone** Type `ping 192.168.1.15` (use the phone's real IP). On Mac/Linux use `ping -c 4 192.168.1.15`.
5	**Read the result** Replies with a time in ms mean the devices can talk. Check the line **Lost = 0 (0% loss)**.

SAMPLE OUTPUT (successful ping, yours will differ)

```
C:\> ping 192.168.1.15
Reply from 192.168.1.15: bytes=32 time=6ms TTL=64
Reply from 192.168.1.15: bytes=32 time=4ms TTL=64
Reply from 192.168.1.15: bytes=32 time=5ms TTL=64
Reply from 192.168.1.15: bytes=32 time=4ms TTL=64
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

If ping fails but both are on Wi-Fi: the phone may be asleep (wake the screen), the phone may block ping replies, or the router may have **AP/client isolation** turned on (common on guest networks). Also check you typed the IP correctly.

My result (phone IP, laptop IP, replies, average time, packets lost):

2 Traceroute to www.youtube.com

1	**Open the terminal** Windows: Command Prompt. Mac/Linux: Terminal.
2	**Run traceroute** Windows: `tracert www.youtube.com`. Mac/Linux: `traceroute www.youtube.com`.
3	**Wait for it to finish** It prints one line per hop. Windows ends with **Trace complete.** and can take up to a minute.
4	**Count the hops** The **hop count is the number on the last line** before it finishes. Write it down.

SAMPLE OUTPUT (shortened; your routers and times will be different)

```
C:\> tracert www.youtube.com

  1     2 ms     2 ms     1 ms  192.168.1.1        (your home router)
  2     9 ms     8 ms    10 ms  10.20.0.1          (ISP network)
  3     *        *        *     Request timed out.
  4    14 ms    13 ms    15 ms  203.0.113.45       (ISP backbone)
  5    19 ms    18 ms    20 ms  198.51.100.7
 ...
 11    24 ms    23 ms    25 ms  142.250.x.x        (Google server)

Trace complete.        <-- last number = 11 hops in this sample
```

How to understand it

What you see	What it means
Hop 1 (usually 192.168.x.x)	Your own home router, the first stop for all your traffic.
Three times (ms) per line	Three test packets. Smaller numbers mean a faster connection.
* * * Request timed out	That router does not answer traceroute. It is still counted as a hop and is not an error.
Last line	The destination (a Google/YouTube server). Its number is your total hops.

Good to know: hop count changes with your ISP, location and time, and usually falls between about 8 and 20. If you see only * * * from some point onward, your network may block traceroute. Try a mobile hotspot to compare.

My result (number of hops and total time to the last hop):

3 Ping a Disconnected Device

1	**Disconnect one device** Turn off Wi-Fi on your friend's phone (or switch the phone to Airplane mode). Keep your laptop connected.
2	**Ping it again** From the laptop run the same command: `ping 192.168.1.15`.
3	**Record the message** Write the exact error text, and the final **Packets** line, in the box below.

SAMPLE OUTPUT A (Windows, device not found on the local network)

```
C:\> ping 192.168.1.15
Reply from 192.168.1.20: Destination host unreachable.
Reply from 192.168.1.20: Destination host unreachable.
Reply from 192.168.1.20: Destination host unreachable.
Reply from 192.168.1.20: Destination host unreachable.
```

SAMPLE OUTPUT B (another common result)

```
C:\> ping 192.168.1.15

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)
```

What the errors mean

Message	Meaning in simple words
Destination host unreachable	Your laptop looked for the device on the network and could not find it. In sample A, 192.168.1.20 is your own laptop reporting the failure.
Request timed out	The ping was sent, but no reply came back in time. The device is off the network, asleep, or ignoring pings.
100% loss	None of the test packets got through, so the devices can no longer communicate.

Tip: on Windows, "Destination host unreachable" can still be counted under *Received* because your own laptop sent that message. Always read the message, not just the count. On Mac/Linux the equivalents are *Request timeout* and *No route to host*.

My result (exact error message, packets sent, received, lost):

4 Device Table with arp -a

1. **Reconnect everything**
   Make sure all your devices (phones, TV, laptop, tablet, console) are on the home Wi-Fi.
2. **Wake the ARP table**
   Ping your router (`ping 192.168.1.1`) and a few devices so they appear in the list.
3. **Run the command**
   In Command Prompt type `arp -a`. Look under the interface that matches your laptop's IP.
4. **Copy the entries**
   Write each **Internet Address** (IP) and **Physical Address** (MAC) into the table below.
5. **Match devices to owners**
   Compare with each device's own IP (Wi-Fi settings), or check your router's "connected devices" page. You can also turn one device's Wi-Fi off and see which entry disappears.

SAMPLE OUTPUT (arp -a)

```
Interface: 192.168.1.20 --- 0x7
  Internet Address      Physical Address      Type
  192.168.1.1           a4-2b-b0-11-22-33     dynamic
  192.168.1.15          3c-5a-b4-aa-bb-cc     dynamic
  192.168.1.255         ff-ff-ff-ff-ff-ff     static   (ignore: broadcast)
  224.0.0.22            01-00-5e-00-00-16     static   (ignore: multicast)
```

My home Wi-Fi device list

IP address	MAC address	Device / owner	How I identified it
192.168.1.1	*a4-2b-b0-11-22-33*	*Home router (example)*	*Gateway; same IP as in tracert hop 1*
192.168.1.15	*3c-5a-b4-aa-bb-cc*	*Friend's phone (example)*	*Matches IP in phone Wi-Fi settings*

Skip these entries: addresses starting with 224. or 239. (multicast) and ones ending in .255 or 255.255.255.255 (broadcast). **Note:** many phones use a *private (random) MAC address* per network, so the MAC may not match the one on the phone's "About" page.

5 One-Page Report

Title: Testing Wi-Fi Connectivity Using ping, traceroute and arp

Complete the blanks with your real results, then add screenshots in the boxes below.

1. Aim

To check whether devices on the same Wi-Fi can communicate, to trace the path to www.youtube.com, and to list the devices on my network.

2. Network setup

I connected my laptop and my friend's phone to the Wi-Fi network named _____. My laptop's IP address was _____ and the phone's IP address was _____. I found them using ipconfig and the phone's Wi-Fi settings.

3. Ping test

I ran `ping` _____ from the laptop. The result was _____ replies with an average time of _____ ms and _____ packets lost, so the devices _____ (could / could not) communicate.

4. Traceroute test

I ran `tracert www.youtube.com`. It took _____ hops to reach the destination, and the total time was about _____ ms. Hop 1 was my home router at _____. Some hops showed * * * because _____.

5. Disconnect test

After turning off Wi-Fi on the phone, I pinged it again and got the message "_____" with _____ packets lost. This shows that _____.

6. Device table

Using `arp -a` I found _____ devices (see the table in Task 4), including my router, laptop, phone and _____.

7. Issues and conclusion

Problems I faced: _____. How I solved them: _____. In conclusion, ping checks if a device is reachable, traceroute shows the path and the number of hops, and arp -a lists the devices and their addresses on the local network.